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TOYOTA GAZOO Racing Europe

Hiring Team

Application: Master Thesis Computer Vision for Motorsport Video Analysis

Dear Hiring Team,

I am applying for the Master Thesis Opportunity on Computer Vision for Motorsport Video Analysis at TOYOTA GAZOO Racing Europe in Köln. My Real Time Flight Tracking Data Pipeline sits closest to the shape of a high frequency motorsport telemetry and video workload. I set up real time collection from the OpenSky Network API, polled live positions every 30 seconds, enriched each against airport, aircraft, and weather data, cleaned raw data with PySpark on Google Cloud, and orchestrated the whole system on Airflow so batch and real time layers refresh automatically every 15 minutes, patterns that carry over cleanly to lap and stint scale video analysis.

My Hybrid RAG Orchestrator shows I can design modular AI systems, with a Llama 3.1 8b core via Groq, a routing agent, ChromaDB with local persistence, and a stateful memory agent shipped behind a Streamlit interface. My CreditIQ project shows I take evaluation rigor, honest reporting, and regulator defensible trade offs as seriously as raw model performance, all useful when a thesis needs a defensible evaluation section.

I am glad to work on site in Köln and my German is at B1 and improving. I would welcome the chance to discuss the thesis.

Kind regards,

Rahul Rawat